Note

We can also begin the Fibonacci sequence with 0 and 1, instead of 1 and 1.

Think

Are two consecutive Fibonacci numbers relatively prime?

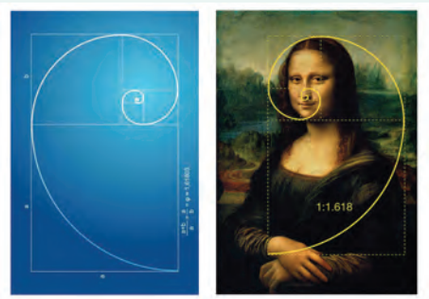
DO YOU KNOW?

Golden Ratio:

Consider the ratio of successive Fibonacci numbers $\frac{3}{2}=1.5$, $\frac{5}{3}=1.66$, $\frac{8}{5}=1.6$, $\frac{13}{8}=1.625$, $\frac{21}{13}=1.6153,\ldots$) you can see the pattern, getting closer to 1.618 and that is denoted by Φ called the Golden Ratio (Φ=1.618). It is observed that shapes having Golden Ratio appear beautiful.

DO YOU KNOW?

The Portrait of Mona Lisa has Fibonacci spiral pattern. This is one of the reasons for the enhanced beauty of Mona Lisa.



5.3 Euclid's game

Ammu and Balu are playing a game. Each one can choose any number and they write it down on a piece of paper. If Ammu picks up the number greater than what Balu picked up, then Ammu will find the difference of the two numbers. That difference will be shown to Balu. Now Balu takes the chance to find the difference between the number what he has and the number shown to him by Ammu. They will continue the process until the difference and the numbers they have become equal. Finally, the person who gets the number equal to the difference wins the game Let us see how it works

Ammu	:	(34,19)	34-19=15
Balu	:	(19,15)	19-15=4
Ammu	:	(15, 4)	15-4=11
Balu	:	(11, 4)	11-4=7
Ammu	:	(7, 4)	7-4=3
Balu	:	(4, 3)	4-3=1
Ammu	:	(3, 1)	3-1=2
Balu	:	(2, 1)	2-1=1
Ammu	:	(1, 1)	same numbers

Suppose Ammu picked the number 34 and Balu picked the number 19. Ammu first finds the difference between 34 and 19 which gives 15. She shows the difference to Balu. Now Balu has 19 and she has 15, the difference is 4. He shows to Ammu and so on. (the bigger number should be kept first to find the difference). So Ammu wins.

E9K7T8

Now suppose they start with Ammu (24, 18).

6th_ Maths _Chapter 5.indd 75 14-10-2019 20:12:51